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Digital Media Design (East China Normal University)

Portfolio for University of Leeds - Design MA Specialism: Digital and Interactive Design

- Creative Computing
- Interaction Design
- Performance art

Designer & creative technologist bridging physical computing, real-time visual systems, and multimodal interaction.



01 Signal Lag

Distributed Storytelling Installation

02 The Post-linguistic Era

Nonverbal Expression | Tactile Instrument | Performance art | Interactive Installation

03 Traces Imprinted

Installation | Interactive Book | Projection Mapping | Tourist Gaze

04 EcoSyntax

Environmental | Speculation | Synthetic Ecosystem | Artificial Biodiversity | Eco-Futurism



THE POST-LINGUISTIC ERA

Nonverbal Expression | Tactile Instrument | Performance Art | Interactive Installation

The Post-Linguistic Era is an interactive sound installation composed of a large number of custom-made stainless steel “gesture plates” and capacitive touch sensors connected to an Arduino system. When touched, each plate triggers a specific synthesized sound through SuperCollider, generating a fluid, ambiguous audio response that resembles the tonal instability of a theremin. As a whole, the installation behaves like a tactile musical instrument, one that users must “play” by physically interacting with symbolic bodily forms, rather than using traditional notes or keys.

This project explores the tension between the desire to communicate and the inevitable misunderstandings caused by the limitations of both language and gesture. By designing an ambiguous nonverbal system based on posture, sound, and dance, I aim to express a deep emotional state of loneliness, one that emerges not from silence, but from being unheard.

Video Link: <https://youtu.be/MjTHjzha6Yo>



SuperCollider



Arduino

INSPIRATION

My Story

I often notice a gap between what I intend to express and what others actually understand. Such discrepancies frequently lead to unnecessary misunderstandings and conflicts. In today's world, social interactions are increasingly mediated through smartphones, cutting off face-to-face connection, making it difficult to convey emotions, tone, or gestures which stripping language of its deeper essence of "communication."

Through this project, I aim to explore the possibilities of non-verbal communication, while also expressing the paradoxical feeling of longing to be understood yet frequently being misunderstood.

DESK RESEARCH

Misconceptions About Nonverbal Communication

✕ People Communicate Using Decodable Body Language.

✕ People Express Emotions Through Universal, Evolved, Iconic, And Categorical Facial Expressions.

Nonverbal communication is often mistaken for a universal "body language," with the belief that gestures, personal space, facial expressions, and deception cues have clear, reliable meanings. In reality, these assumptions are misleading.

Unlike language, nonverbal signals lack propositions, vocabulary, and grammar, making them far less precise or efficient. They offer only partial clues rather than exact meanings, leaving much open to interpretation. As a result, nonverbal communication is inherently ambiguous and easily misunderstood.

Miming

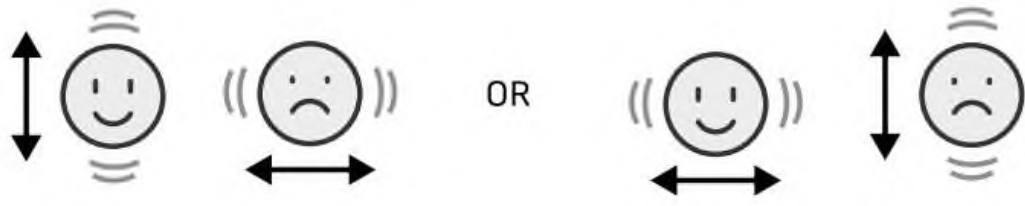
Charlie Chaplin



A pioneering English comic actor, filmmaker, and composer of the silent film era, best known for his "tramp" character who marked by quirky gestures, shabby yet tidy clothing, puppet-like movements, and meticulous grooming.

Misunderstandings In Cross-Cultural Nonverbal Communication

Nod And Shake

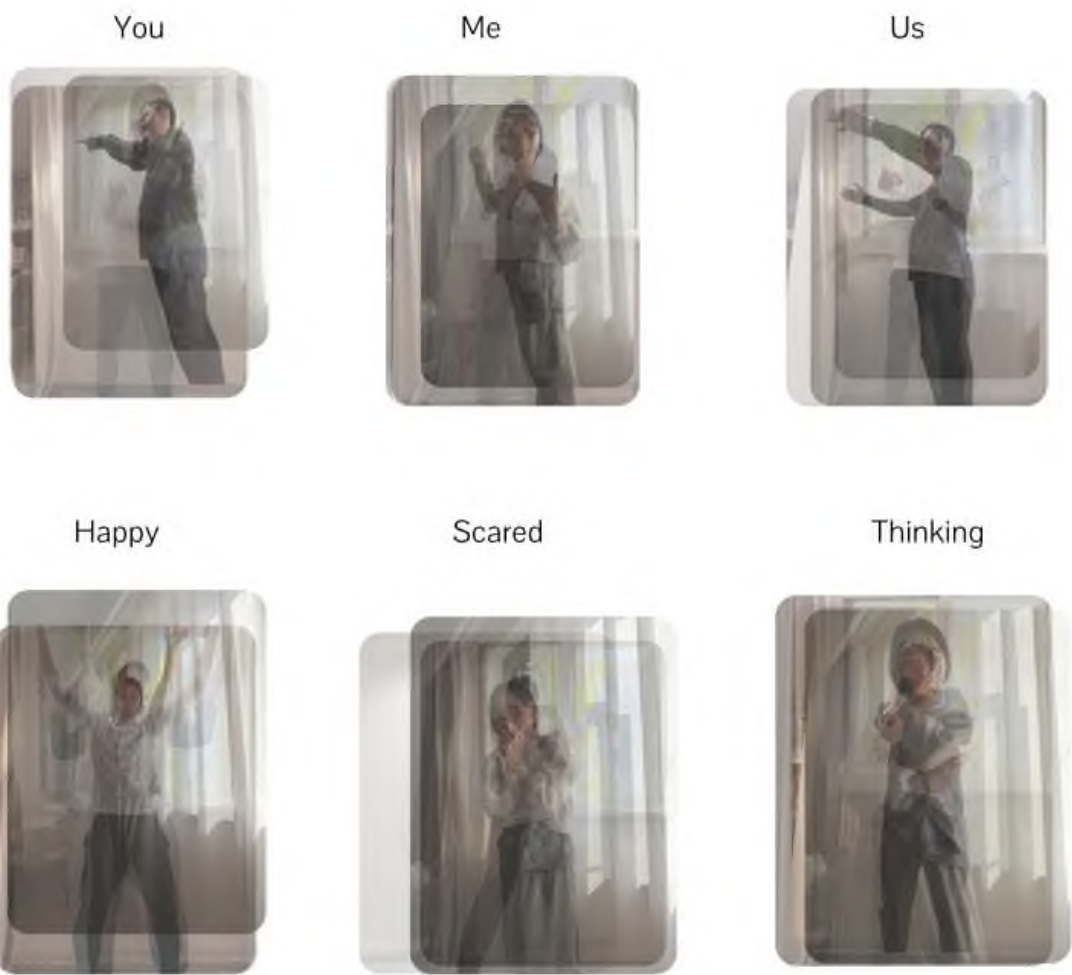


Head gestures can convey different meanings across cultures. Nodding generally signifies "yes" or agreement in most cultures, but in Greece, Cyprus and Iran, it means "no."

EXPERIMENT OF SHAPE

Practical Research

I provided the participants with a set of words, such as verbs, adjectives, nouns, and pronouns, that are commonly used in verbal communication. They were then asked to express these words solely through body language, in order to explore the similarities and differences in nonverbal expression among individuals.



To summarize the meanings represented by different body postures, I posing a puppet and assigning to each posture the different interpretations I understood.



Conclusion



Although participants' postures were largely similar, the differing interpretations of specific gestures show that nonverbal expression has limited universality and offers little long-term feasibility as a standalone communication system.

THE FORMATION OF SYMBOLS

Artist Reference

Arrival-Heptapod Languages

This nonlinear and symbolic form of expression influenced my exploration of emotional posture symbols.



Attempts

At first, I attempted to express emotions directly through patterns of varying shapes. Later, I further simplified the structure of the body by using a pair of curves to represent the limbs, while reducing the head and waist to a single point.



I experimented with mannequin poses to imagine variations of bodily expression. I eventually created 65 symbols, each functioning as part of a self-invented nonverbal vocabulary that form the foundation of my visual language.



Extract The Curve Contour

However, I found the symbols lacking aesthetic refinement. I then used continuous lines to outline their overall forms, making them more abstract and symbolic. However, these shapes turned out to be difficult to distinguish from one another.



This version serves as the draft for the final design. I continued using curved lines, representing both arms and legs with a single stroke. The postures remain recognizable.

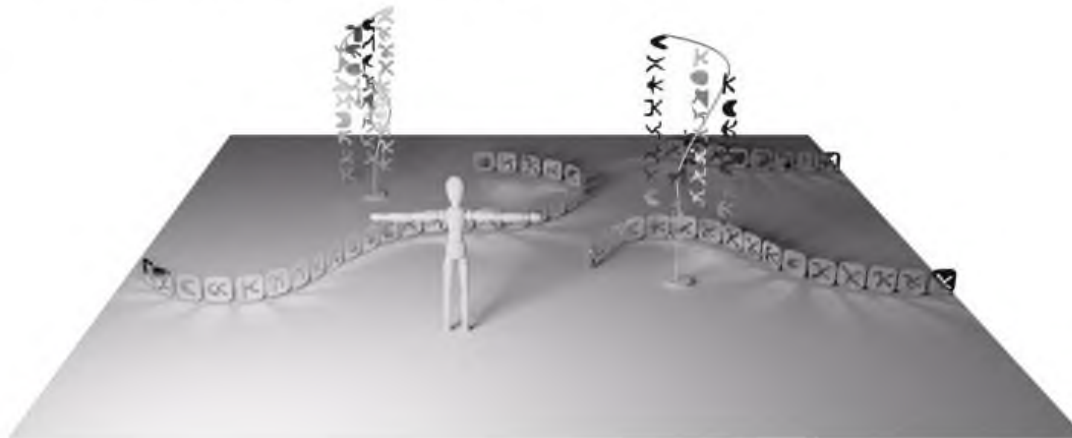


3d Modeling

I visualized the "posture symbols" through 3D modeling in C4D, experimenting with various modeling approaches.



During the rendering stage, I chose a metallic material. The cold and distant qualities align with my exploration of the emotional isolation.



ATTEMPTS AT INSTALLATION FORMS

Version Three: Three Dimensional Printing

I reconstructed the symbol shapes in C4D and experimented with 3D printing to produce physical samples. I printed two versions: a thin-plate version and a three-dimensional version. The first version produced better results and preserved the clarity of the lines.



Building on this, I came up with the idea of adding small holes along the edges of the thin plates so that each piece could be linked with fine chains like individual words forming a sentence.



Version Four: Laser Printing On Stainless Steel Sheets

I eventually decided to use laser-cut stainless steel for fabrication. Initially, I experimented with 3mm-thick sheets, but the overall weight became excessive and impractical for installation and display. After several tests, I settled on a 0.5mm thickness, which maintained both structural stability and a sense of lightness.

During the process, I also noticed that some patterns had overly sharp edges, posing potential safety concerns, so I refined the design by softening those contours.



I suddenly realized that by cutting out the negative shapes of the symbols and linking them together, I could create freestanding sculptures that stand on the floor or tabletop. This introduced a new form of presentation which is transforming the once-flat symbols into spatial expressions of language.



FIELD DESIGN CONCEPT

Part1 Visual

I sketched a preliminary layout of the installation on paper: the longer curves represent stainless steel structures, while the shorter lines symbolize suspended elements.



Part2 Sound - Theremin

I was inspired by the Theremin, an instrument that produces sound through bodily interaction. Often described as “a voice from the cosmos,” its trembling, distant tone evokes solitude. I incorporated this idea into my installation, using sound and bodily interaction as another form of communication.



Part3 Dance

Body movement became a live component, extending the installation into spatial communication.



DESIGN PROCESS

Visual Effect

In constructing the installation, I used thin metal chains and two sizes of metal rings, totaling eight 1-meter chains and over 400 connecting rings.



The larger rings were used to link the hollow stainless-steel plates, ensuring that they could stand stably on the ground and maintain structural balance.



Additionally, I discovered a flexible PVC tube that can be freely bent, and I purchased two lengths, 5 meters and 2 meters, for suspending groups of metal plates.



Sound

I used SuperCollider to create both the background music and the interactive sound triggered by touch, experimenting with tones reminiscent of the Theremin.

The sound carried an ethereal quality, hovering between presence and absence. This ambiguous and fragile sonic atmosphere resonates with the central themes of my work: the solitude of communication and the elusive nature of understanding.

The Code For Background Music



The Code For The Keyboard Interaction Sound



Access Hardware

To enable sound activation through direct touch on the installation rather than the computer keyboard, I used a MPRI21-Breakout-v12 capacitive touch sensor connected to an Arduino board.



The Code For Touch Interaction



Each sensor has 12 available input pins, allowing me to connect a total of 24 touch points that correspond to different tones or sound fragments.



When a viewer touches a metal plate (use conductive glue to attach copper wires), the capacitive signal is detected by the sensor and sent through the Arduino to SuperCollider, triggering the corresponding sound.



SCENE SETUP

For the venue, I was able to rent a badminton court to set up the installation and complete the shoot.



We initially tried a dim lighting setup but failed to clearly capture the installation, so we covered the floor with white fabric to reflect more light. I also used a smoke machine to add a layer of hazy atmosphere to the overall effect.

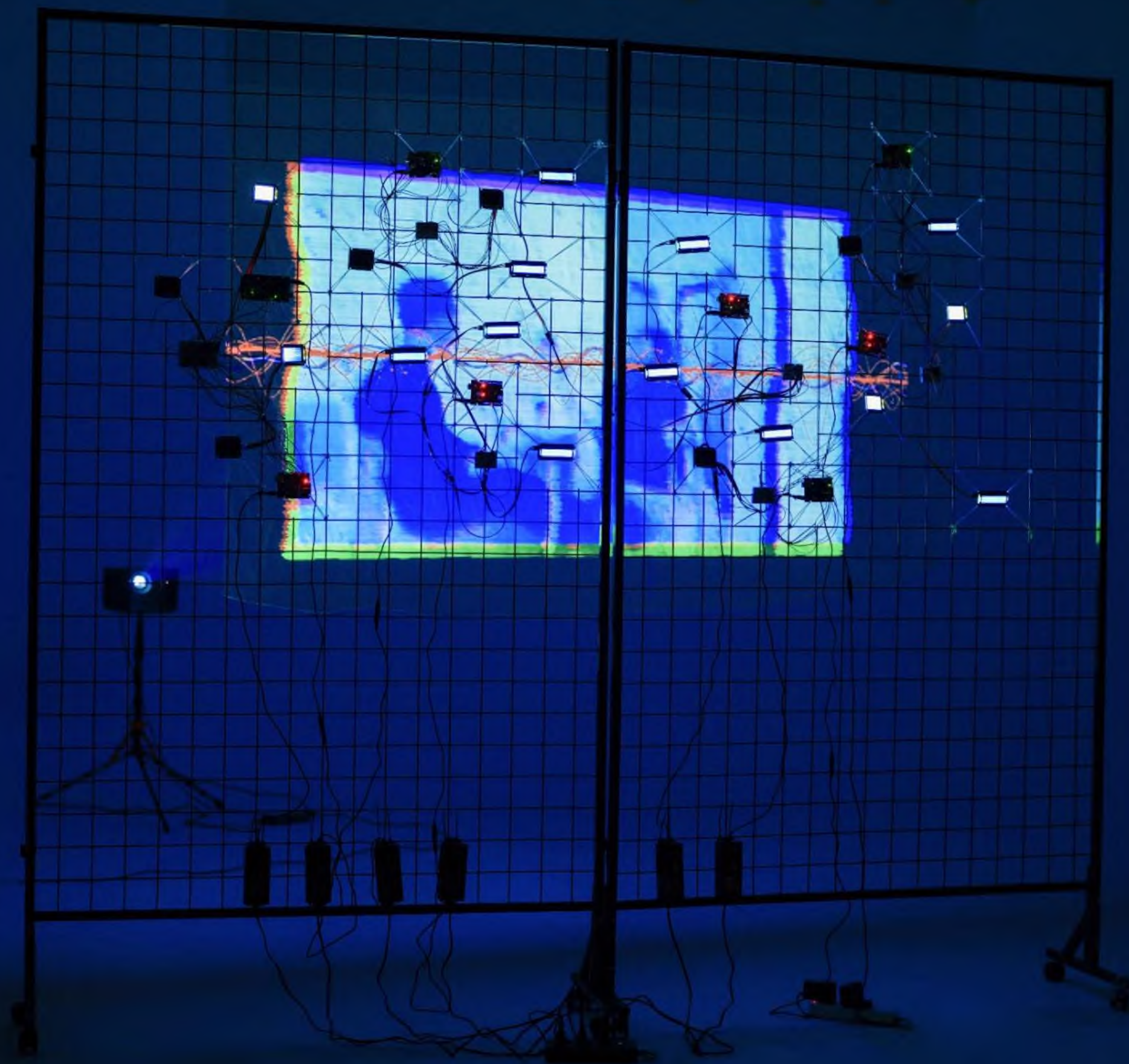






Video Link: <https://youtu.be/MjTHjzha6Yo>





Signal Lag

Installation, Embedded Electronics, Emotional Asymmetry, Controlled Speech

Signal Lag is a media installation combining a grid-based screen array and a rear-projected experimental video. LCD character displays and small TFT modules are mounted onto a single steel mesh frame, through which fragmented police task messages and delayed family replies emerge asynchronously. Behind the mesh, a translucent screen captures color-treated interview footage, creating interference between presence and obscurity, intimacy and system. Originally developed from interviews with families of police officers, the project explores how institutional structures, emotional suppression, confidentiality, erratic schedules—induce signal delay and narrative rupture between law enforcement personnel and their loved ones. Police officers are trained to withhold vulnerability, yet this very silence amplifies the emotional labor of those waiting at home.

The installation's broken text rhythm and low-resolution display aesthetics resonate with the fragility of connection. Combined with disrupted moving images, **Signal Lag** embodies the emotional architecture of distance, protocol, and love strained through layers of delay.

Video Link: <https://youtu.be/PzeDgDzhNWI>

INSPIRATION

> MY STORY

Both of my parents are in the police force, and my boyfriend is studying at the Shanghai Public Security College. Therefore, I have a certain understanding of the police profession and what is it like being a **police officer's family member**. When I was a child, my parents were often too busy with work to take care of me, and I was almost raised by my grandmother alone. Due to the special nature of their work, they often **couldn't reunite with the family** during holidays because of overtime or other tasks. However, over time, it becomes a habit.



After I started dating my boyfriend, I often couldn't see him because of his busy internship during holidays. I have to **got used to his schedule** and tried to spend more time together when he was free.

> NEEDS OF ATTENTION

But such a habit **does not mean the disappearance of contradictions**, it hides behind passivity, helplessness and self-sacrifice.

Such family problems are transparent in the society, so I decided to find some similar families and dig out the bitterness and helplessness behind each of their families, and let such **feelings be seen** by everyone.



DESK RESEARCH

> POLICE WORK AND FAMILY BALANCE

Due to the unique nature of the profession, police officers' family members must be compelled to **passively shoulder additional psychological and emotional burdens** inherent to the law enforcement profession.

>> MAJOR SOURCE OF STRESS FOR THE POLICE

ORGANIZATIONAL

Hierarchies
Shifts
Inadequate resources
Policy restrictions

OCCUPATIONAL

Frequent exposure to major incidents and human suffering

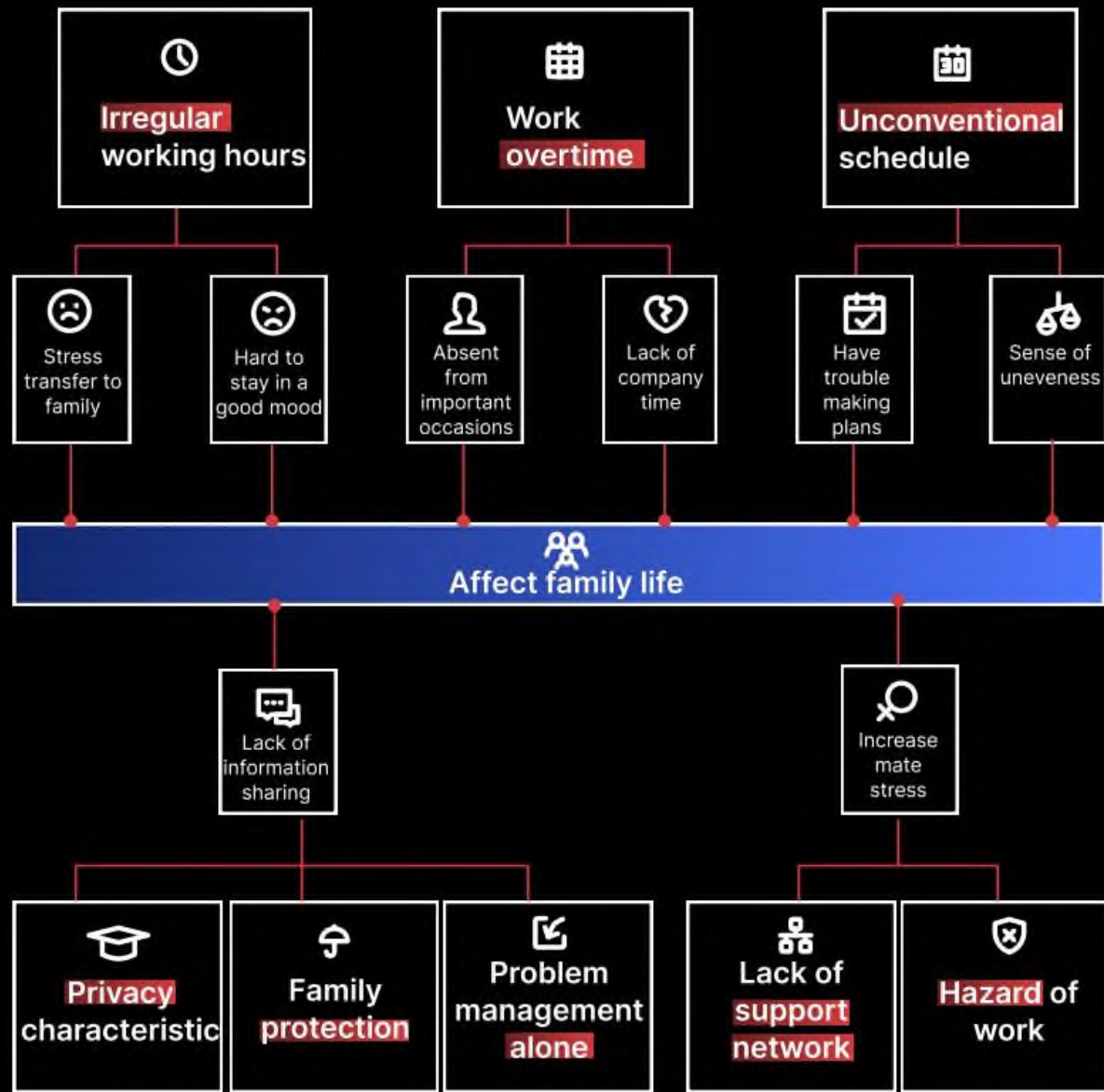
EXTERNAL

Public supervision
Public opinion

PERSONAL

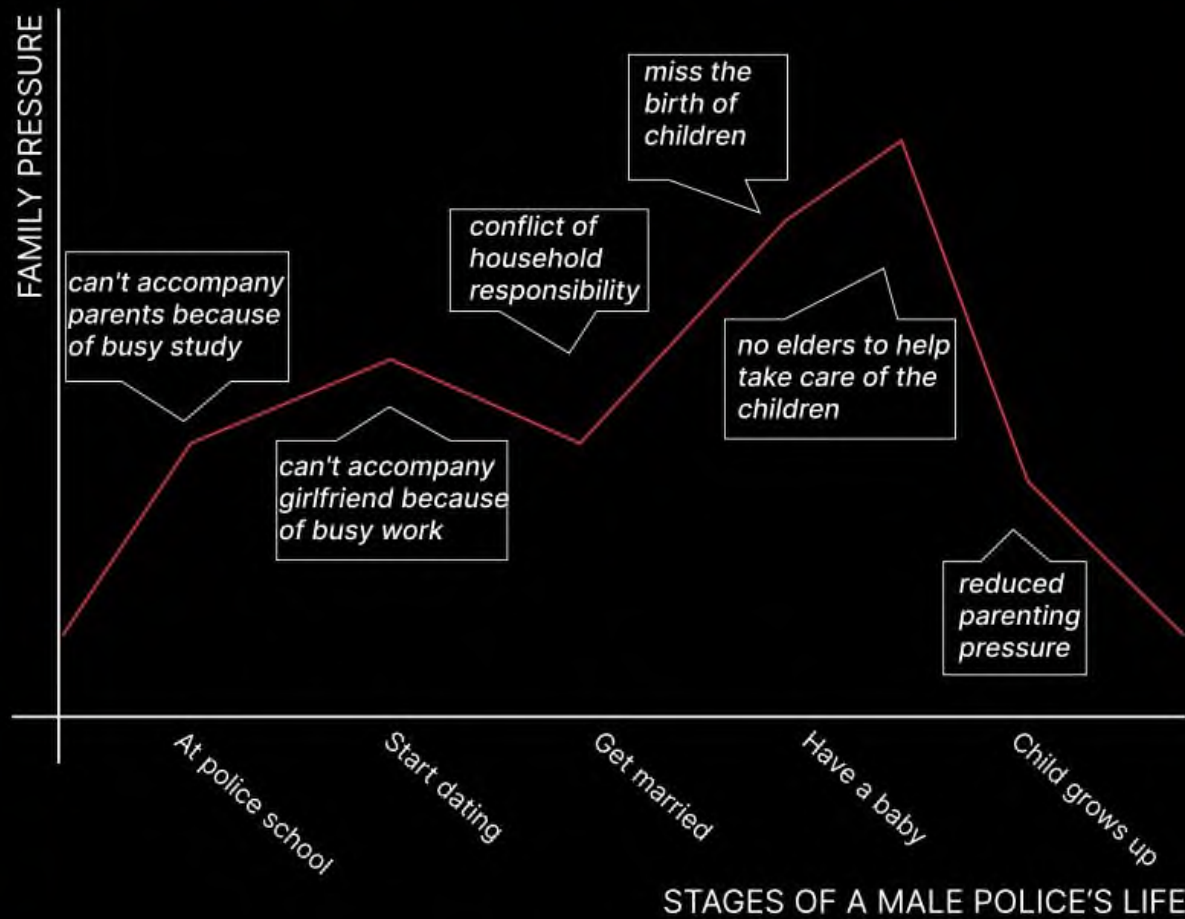
24/7 availability
Absence from family events
Work-family conflicts

>> FACTORS AFFECTING THE FAMILY



<https://www.policecare.org.uk/help/stress/>
<https://link.springer.com/article/10.1007/s11896-023-09619-w>

> CHANGES IN THE IMPACT ON THE FAMILY



INTERVIEW

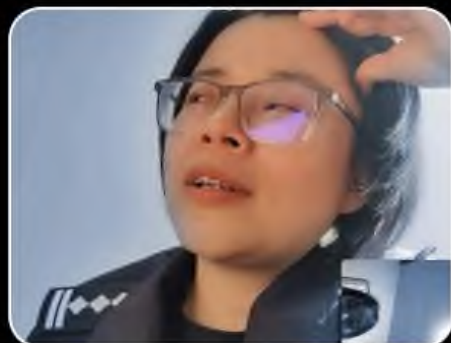
> THE DAILY LIFE OF THE POLICE STATION

With a friend's help, I conducted a one-day field visit to the **Meiyuan Police Station in Shanghai**. The public area has a service window staffed by two officers who handle residents' inquiries. Most daily cases involve minor civil or neighborhood disputes, which require repeated **mediation**. As a result, the station's routine is marked by a constant flow of tasks that keep officers **mentally strained and busy**.



> STORY OF 4 FAMILY

I. ROUTINE BEYOND REUNION



>> MS. WU 47

- Spouse of Detective team member
- Station instructor

>> STORY

Both parents are police officers whose mismatched shifts erase dinners, weekends, and time with their daughter. She once wrote, "Mom and Dad help others reunite, so I stay alone." The parents felt proud, but beneath the pride lay the quiet loss of time, presence, and softness.



II. INTIMACY INTERRUPTED



>> LUCY 24

- Girlfriend of Community officer
- Graduate student

>> STORY

She entered the relationship early, only to face rotating shifts, distance, and his family's hostility; lacking care and protection, she learned silence, carrying the scar alone.



III. SILENCE BETWEEN ORDERS



>> MRS.WANG 54

- Mother of Local precinct officer
- Accountant

>> STORY

Her police-officer son shares little, yet they maintain a quiet bond expressed through small actions and unspoken understanding.



IV. ORDERS, REWRITTEN



>> MS.ZHANG 21

- Daughter of Immigration & Border Inspection
- University student

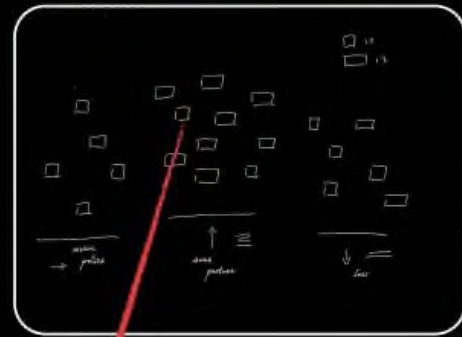
>> STORY

Raised by often-absent border-police parents, she grew independent early, learning that behind their heroism is a long, quiet absence that shapes a child's idea of home.



DESIGN CONCEPT

> MAIN IDEA



I envision designing a wall filled with **display screens**, each silently narrating the personal stories of four police family members through text. Behind these screens, **experimental video** projections will play in layered motion.



The square-shaped screens display the messages received by the police, both in form and function inspired by official police terminals.

> INSTALLATION DESIGN



Two types of screens alternate throughout the installation, connected by a timeline that threads the entire narrative.

The color BLUE Displays **official task orders** received by police on duty

The color RED Shows fragmented messages sent and received by their family members via mobile phones.

>> CONTENT(FOUR STORIES)

[illegible]

These texts capture both the everyday moments and the quiet compromises of their lives. Viewers can sense the **emotional weight** carried by police families, and feel the struggle and helplessness in trying to **balance duty and personal life**.

> EXPERIMENTAL VIDEO DESIGN

I used After Effects and TouchDesigner to give each interview a detached visual treatment to reflect the different psychological states within police families.



>> AUDIO



In addition, I used **Processing** to **visualize** selected audio excerpts from the interviews. These reactive graphics allow viewers to perceive the **emotional rise and fall** of each narrative through movement and form.

DESIGN PROCESS

> VIDEO COMPOSITION

I overlaid the visualized audio lines across the top of the frame.

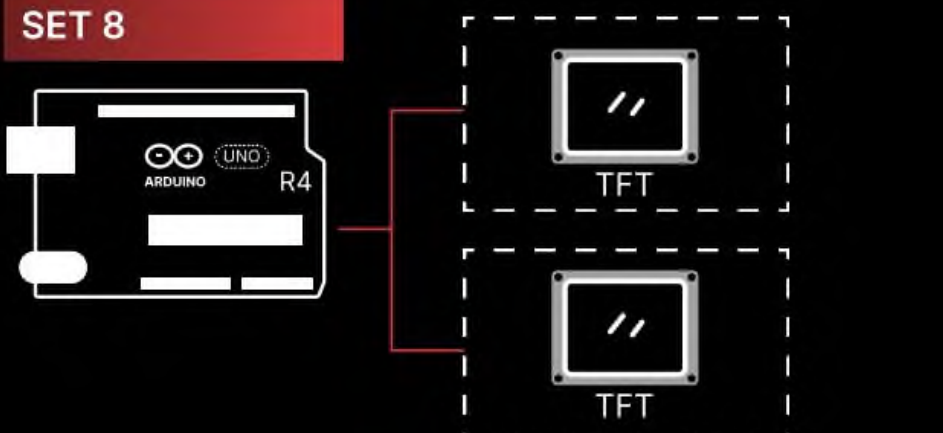
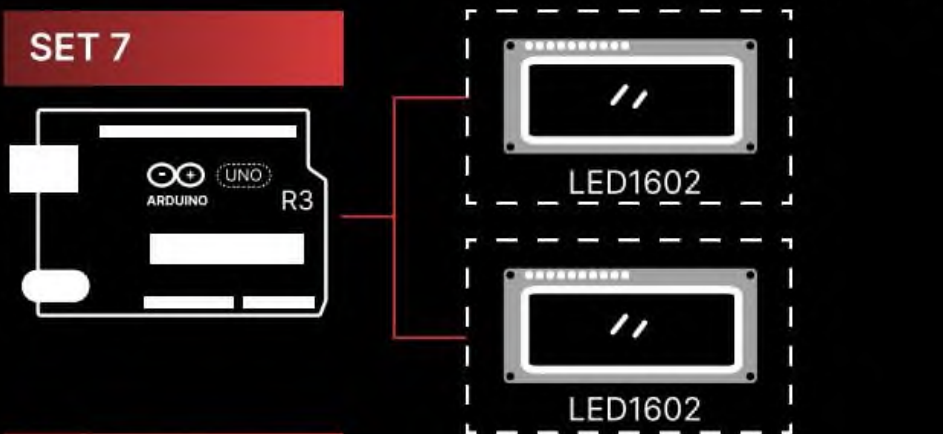
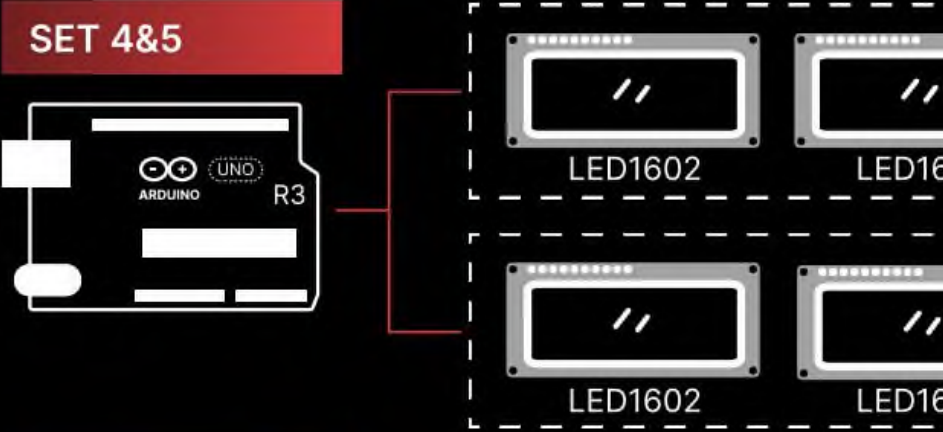
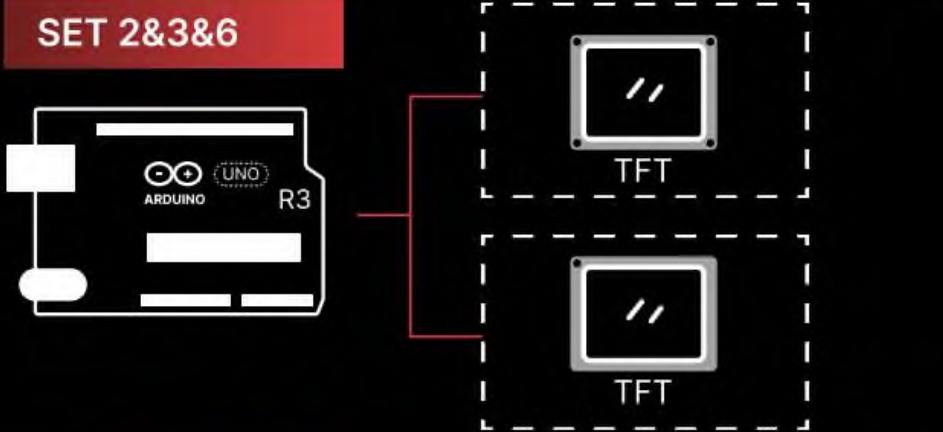
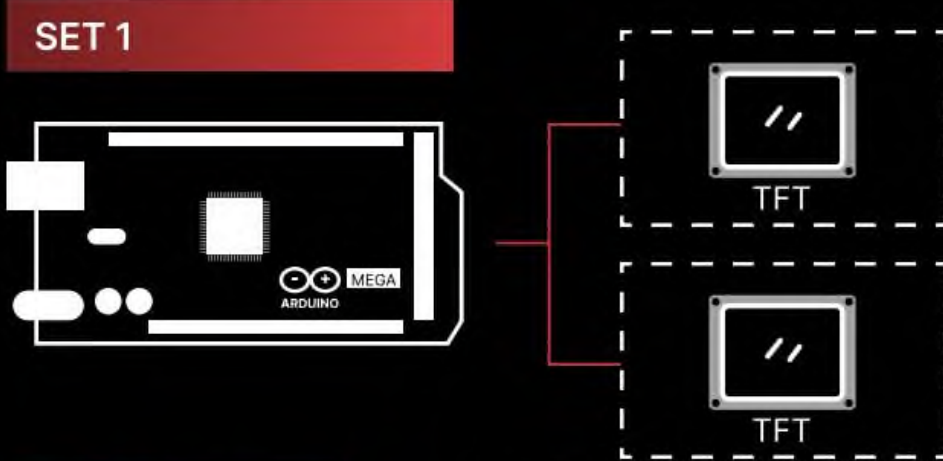


I applied glitch and ripple effects to **simulate the look of an old, unstable television signal.**

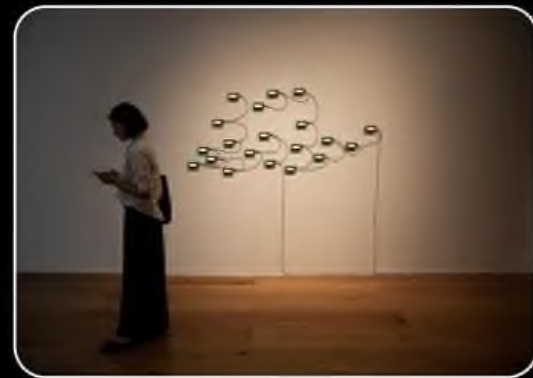
Additionally, I introduced **chromatic aberration effects** along the edges of the frame that generate a sense of **disorientation and emotional distance**.

> HARDWARE CONNECTION

After research, we confirm Arduino to control the screens, with planned quantities and connections using serial or I²C for stable communication.



>> REFERENCE



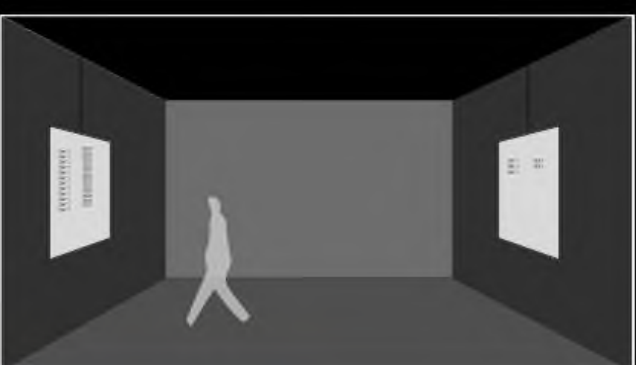
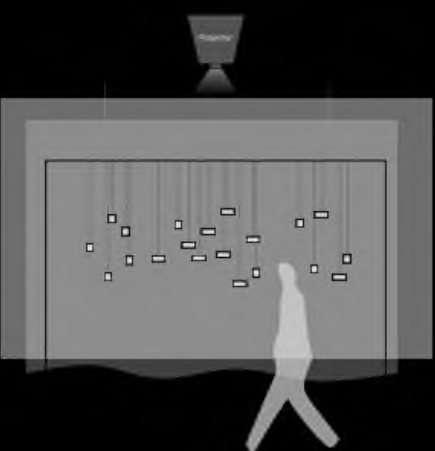
Rafael Lozano-Hemmer, "33 Questions per Minute, Relational Architecture 5", 2000.



>> INSTALLATION SETUP CONCEPT

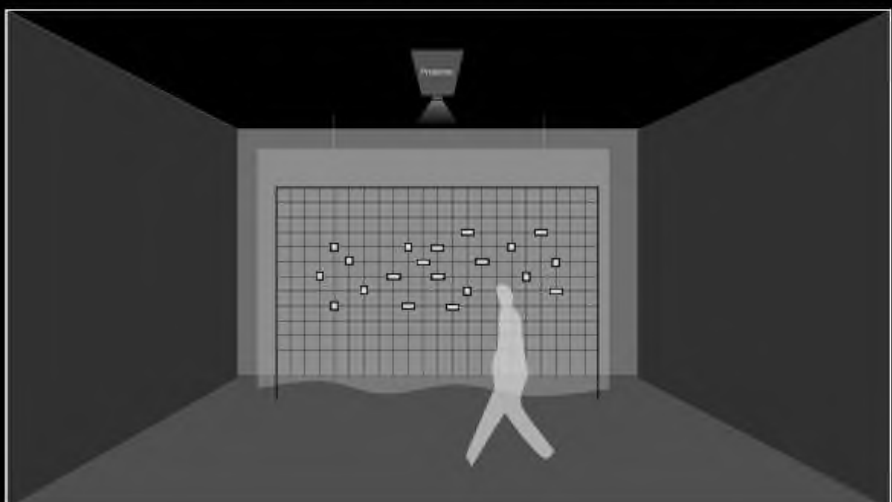
ROUGH IDEA

During brainstorming, I considered several installation approaches: using two **large screens** to separately display messages received by police officers and their families; placing the screens on **opposite sides** of the space for a more immersive experience; or **suspending multiple small screens** on thin wires, each showing a brief message.



FINAL VERSION

I ultimately decided to mount the screens onto a black **grid frame**, ensuring structural stability while maintaining visual layering and aesthetic order.



> CODING



SET 1

SET 2&3&6&8

SET 4&5

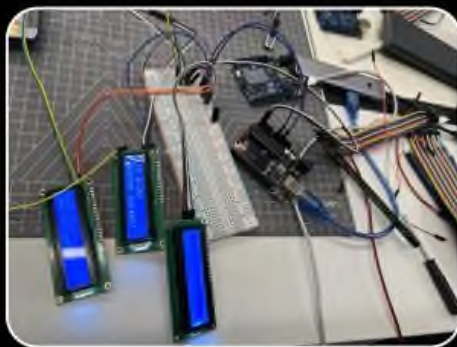


SET 7



> BUILDING

A technical glitch once caused horizontal and vertical **stripes** to appear on the TFT screens. However, this unintended effect resembled **the aged look** of older police equipment, so we chose to retain it as part of the installation's visual language.



>> STEP 0.5 PURCHASING

The main hardware included 6 to 7 Arduino boards, 10 LED displays, 10 TFT screens, multiple jumper wires, 4 to 5 breadboards, fishing line, and several 9V power supplies.

>> STEP 1 DETERMINE&FIX POSITION



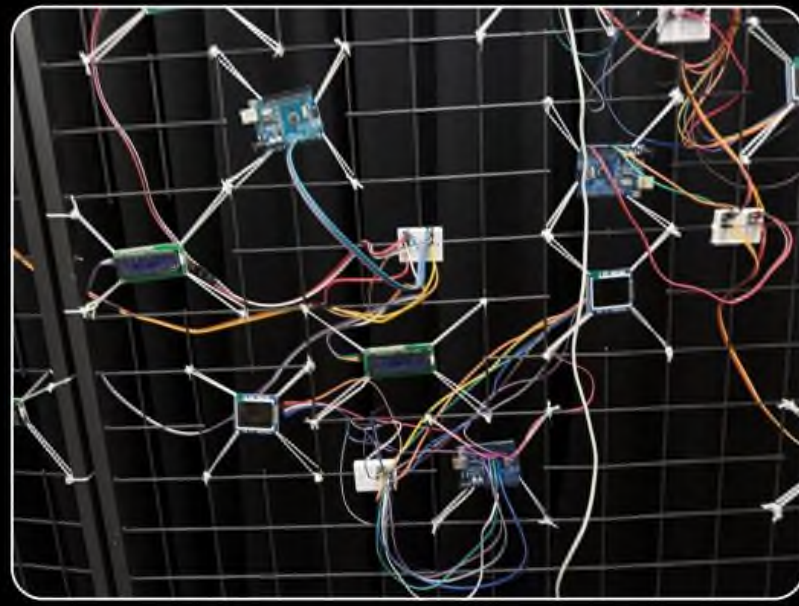
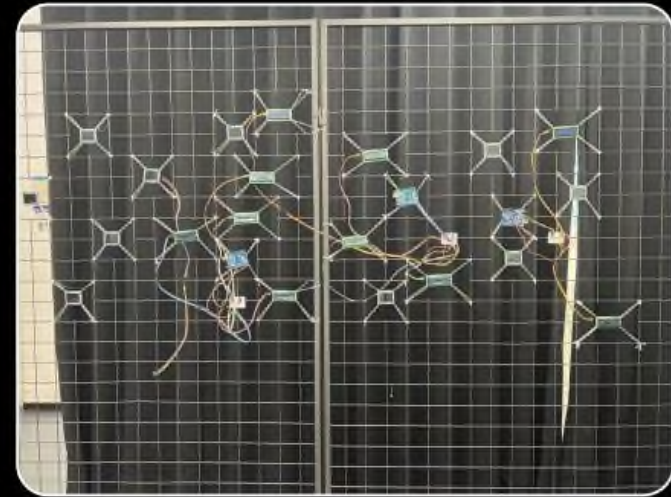
I initially attempted to suspend the screens with fishing line, but its slippery surface made it difficult to secure. I later switched to **cotton thread**, which provided better stability and was easier to work with.

I removed the metal frames around the screens and tied cords to the **four corners** for suspension. This not only improved the visual aesthetics but also symbolized the sense of constraint and tension between police officers and their families.

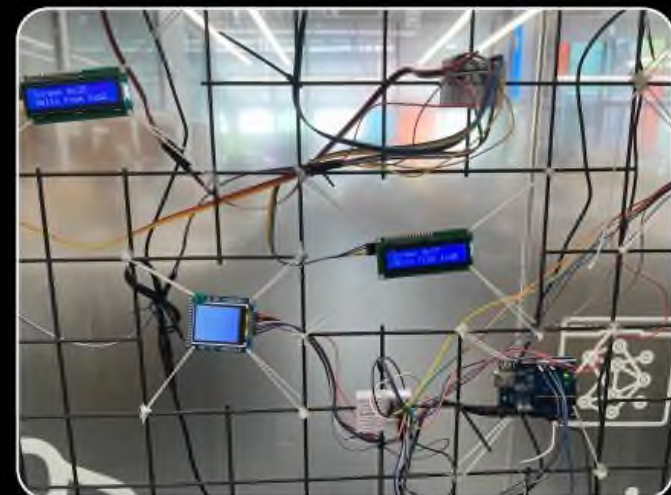


>> STEP 2 WIRING

After securing the screen positions, we connected the Arduino and breadboard to begin **wiring and testing**, preparing the setup for content display.



>> STEP 3 PROGRAM UPLOADING



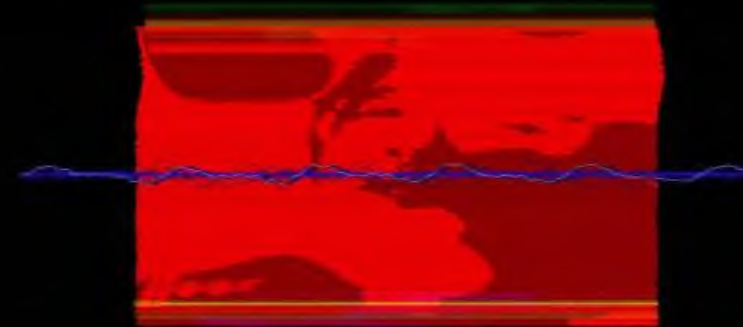
The final step was **uploading the programmed** code to each Arduino board and **powering the system** to test whether all screens displayed correctly.



Frequent issues like short circuits and loose connections occurred during the process, but we resolved them one by one. Finally, we secured all wiring with hot glue to ensure stable operation.

| EXPERIMENTAL VIDEO | REFLECTION

These four experimental videos, drawn from interviews with police families, reveal lives shaped by **pressure and absence, blending pain, quiet pride, and comfort** to portray the complex humanity behind the uniform.



His essay was called uh, he wrote it on the fifteenth day of the first lunar month



I think I'm okay



His dad said you beat him and he said his son



That must have been frustrating

> FUTURE INFLUENCE

Inspired by the recent British drama A Confession, I have gained some insights into **the rules and regulations within** the police system. There are no absolutely good or entirely correct rules in the world, and both police officers and their families are inevitably affected to varying degrees.

Nevertheless, through my work, I hope to help people **better understand the hardships faced by officers and their families**, and to subtly encourage change over time.



> FUTURE WORKS

Due to technical and budget constraints, both the number of wires and Arduino boards used **exceeded initial expectations**, making wiring and debugging more challenging.

Originally, I hoped to install this work **in a high-traffic outdoor location, such as a public bulletin board or campus plaza**, in order to reach a broader audience and spark wider reflection. However, two key limitations prevented this from happening.

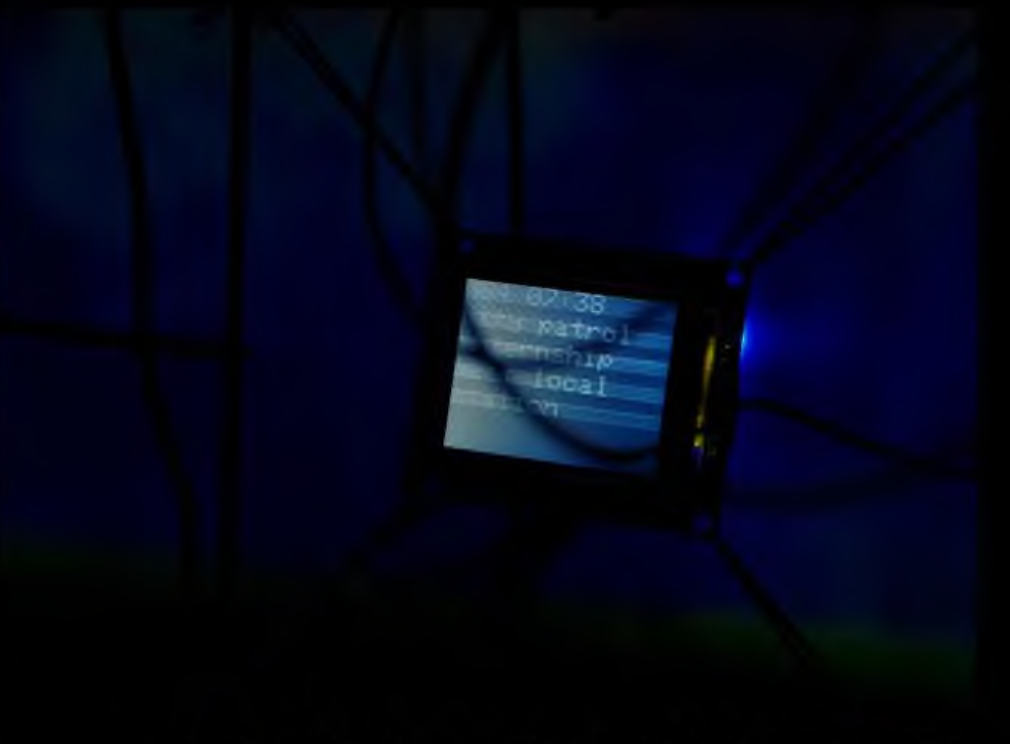
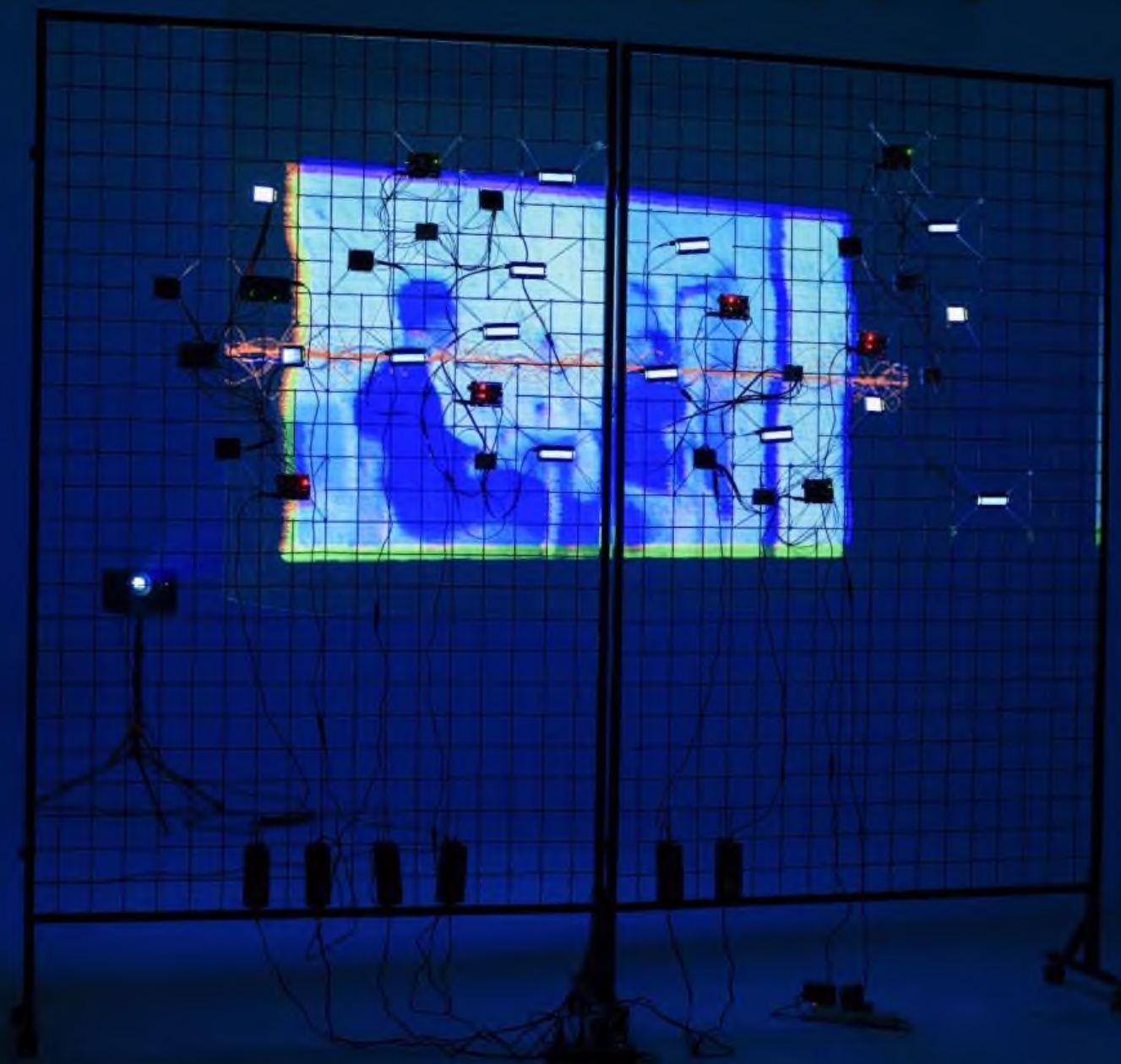
REASON 1

My **technical skills** were not yet developed enough to ensure the piece could operate reliably in an uncontrolled outdoor environment.

REASON 2

As a student, I lacked **the authority and access** to navigate formal approval processes required for displaying work in public spaces.

FINAL - INSTALLATION





Traces Imprinted

Installation, Interactive Book, Projection Mapping, Tourist Gaze

Traces Imprinted is a mixed-media installation combining a hand-bound experimental book, an iPad-based narrative display, newsprint-style posters, and a live camera-feed projection. The book replaces the physical landmark with a readable object. Its pages weaving together site history and past events while visitors are invited to leave inked fingerprints before turning each page, slowly inscribing their presence into the work.

A first-person text channels the “voice” of the site itself, reflecting back on being observed, turning the act of looking into a reciprocal exchange. The surrounding posters borrow the visual language of global news coverage, chronicling the ecological and cultural pressures that mass tourism exerts on destinations worldwide.

The live camera projection overlays real-time visitor movement with the exhibition space, underlining how simply entering a site leaves an imprint, often unnoticed, on its environment. Together, these components invert the typical tourist dynamic: the viewer becomes both the observer and the observed, implicated in the mutual gaze between people and place.

Emerging from research on the “Wu Kang Road bow balcony incident” and broader critiques of overtourism, Tourist’s Gaze questions the pursuit of iconic imagery and the commodification of experience, inviting reflection on the act of “watching” itself.

Video Link : <https://youtu.be/FsKSM0n3gdU?si=XECPSV0yPtmVSi38>

INSPIRATION

The Butterfly Balcony on Wukang Road, Shanghai



In 2021, a balcony on Shanghai’s Wukang Road decorated with a pink ribbon attracted crowds seeking “romantic” photos, **disrupting the daily life of the elderly resident and even local traffic**. Knowing the area well, I chose Wukang Road as the **focus of my project to encourage tourists to reflect on their actions**.

RESEARCH



The Tourist Gaze 3.0



The “Gaze” refers to the way tourists view and appreciate attractions. This act is **shaped by social culture, media, and history**.

Tourists carry **specific expectations and cultural imaginations**, searching for symbols and meanings in different places.



The “Gaze” is a **form of power**, it defines what is considered “worth seeing.”

Tourism and tourist behavior are **reshaping the world**, and in turn, these changes also **influence how tourists behave**.

REFLECTION

This book made me realize that tourists continually **shape the places they visit**. It inspired me to create an **artificial “tourist attraction”** where visitors can freely explore and leave their own traces to prompt reflection on their behavior as tourists.

Theoretical Research— Behavior Settings

Each setting has **typical behavior rules** (norms and expectations).For example: When people are in the classroom, they will voluntarily sit down to study.



What is the Psychology Behind Tourists’ Deliberate Misconduct?

Unchecked bad behavior encourages imitation and lets tourists justify small acts of transgression, which they use to gain attention, mark “I was here,” and symbolically claim the place as their own.

BROKEN WINDOW EFFECT

SOCIAL RECOGNITION

POSSESSIVENESS

ENTERTAINMENT PSYCHOLOGY

REFLECTION

I believe the “attraction” I create **should appear already “damaged” or “polluted,”** so that visitors may develop the mentality of “Even if I contribute to its destruction, others have done it too.” This would reveal how people tend to justify their actions and **avoid responsibility when part of a collective behavior**.

DESIGN CONCEPT

Main Concept

Using the “Wukang Road Bowknot Balcony Incident” as a starting point, this project builds a **critical narrative landscape** through an experimental book, text video, news-style posters, and interactive projection. It critiques **the tourist gaze on popular sites while also reflecting the gaze back on tourists**.

Rather than digitally “solving” overcrowding, it highlights the **symbolic consumption** of places and **prompts reflection** on the act of viewing itself.

book

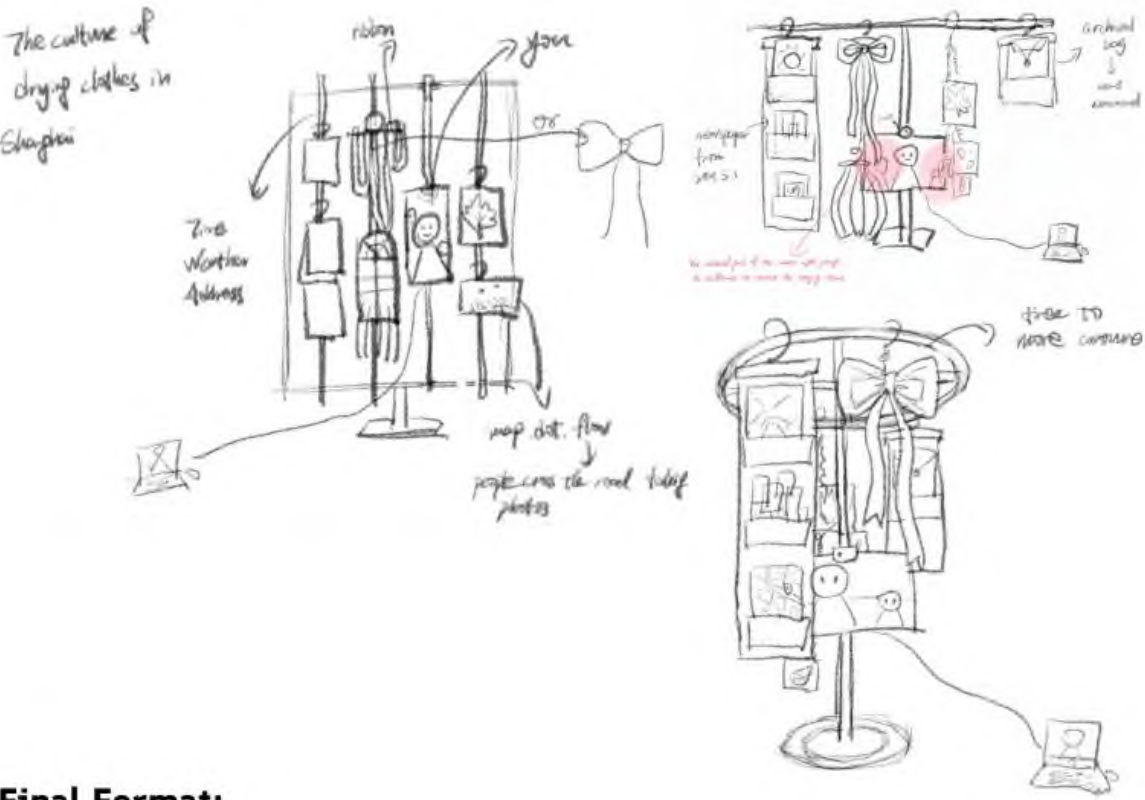
projection

poster

video

Display Format

Abandoned Plan:



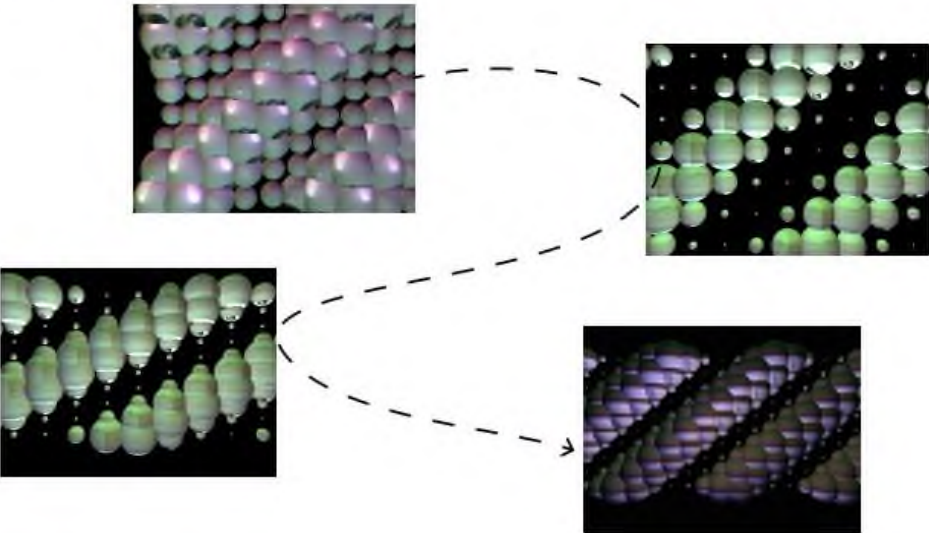
Final Format:



DESIGN PROCESS

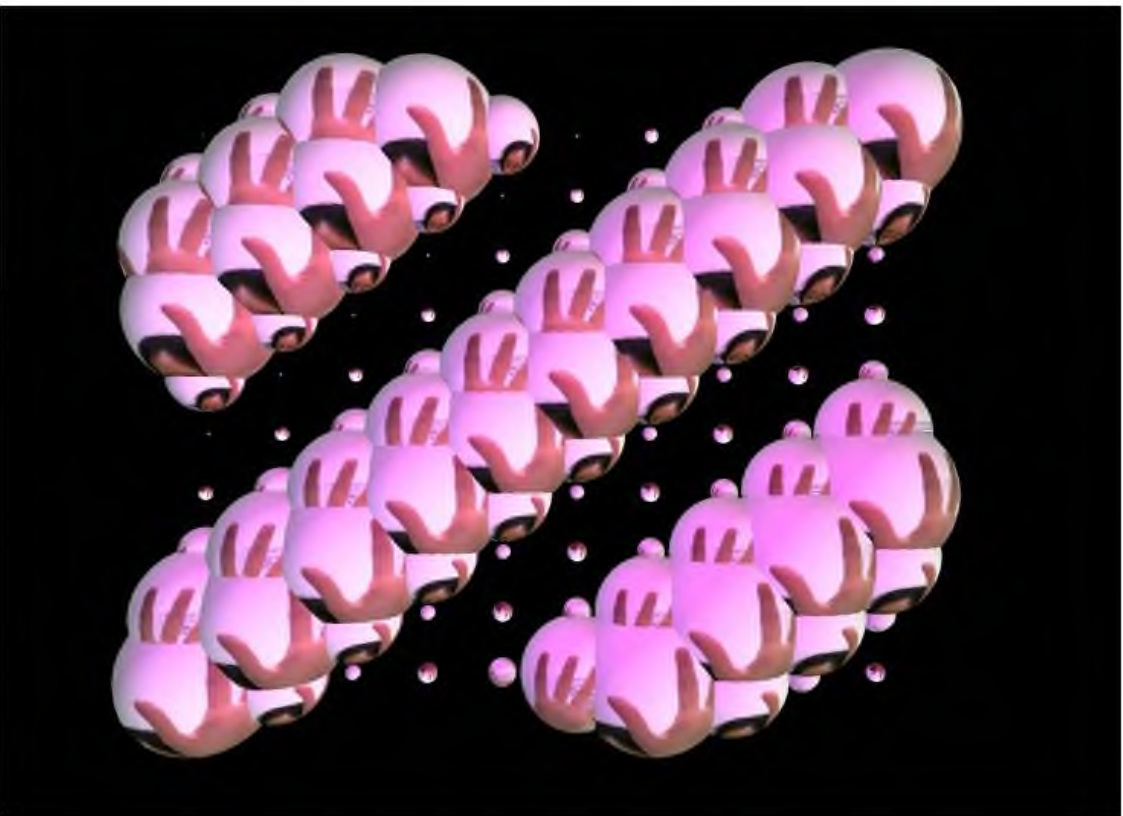
Camera Interaction

Iteration:



Final Effect:

I chose to experiment with this special 3D camera as the core form. The circular spheres around it keep rotating, and whenever someone faces the lens, the camera continuously captures their face or body. As these bubble-like images accumulate, they create the effect of countless viewers watching the scene at once, visualizing the collective “gaze” of tourists.



p5.js Code Implementation in P5.js

```
1 let colors;
2 let capture;
3 let lapse = 0;
4
5 function setup() {
6   colors = "9046cf-cc59d2-f487b6-fff3f0-fde12d".split("-").map(c => "#" + c);
7
8   createCanvas(1112, 834, WEBGL);
9   background(100);
10
11   capture = createCapture(VIDEO);
12   capture.size(320, 240);
13   capture.hide();
14 }
15
16 function draw() {
17   background(0);
18   orbitControl();
19   let ww = 70;
20
21   ambientLight(200);
22   pointLight(255, 0, 0, mouseX - width / 2, mouseY - height / 2, 300);
23
24   pointLight(0, 0, 255, -200, 0, 300);
25   directionalLight(250, 250, 250, -1, -1, -1);
26
27   camera(0, 0, sin(frameCount / 50) * 300 + 800, 0, 0, 0, 0, 1, 0);
28
29   for (let i = -5; i < 5; i++) {
30     for (let o = -5; o < 5; o++) {
31       push();
32       translate(i * (ww + 80), o * (ww + 50), -noise(i / 5, o / 5, frameCount / 40) * 100 - 500);
33       rotateY(i + o + frameCount / 100);
34
35       texture(capture);
36       noStroke();
37       sphere(ww + sin(frameCount / 20 + i + o) * 90);
38       pop();
39     }
40   }
41 }
42
43 function mousePressed() {
44   if (millis() - lapse > 400) {
45     save("img_" + month() + "-" + day() + "_" + hour() + "-" + minute() + "-" + second() + ".jpg");
46     lapse = millis();
47   }
48 }
```

OUTPUT-BOOK

Printed Version

I approached the book design in a rather experimental manner. Since I am not skilled at drawing, I chose to work with collected images through the technique of collage. The cover features a hand-drawn bow, serving as a symbol of tourist gaze, and its color also defines the book's primary color palette.



Manual Processing of Books

Inspired by Leo Virieu's work Garage, I hope that the audience, as "tourists", can **leave behind tangible traces** while reading the book, such as fingerprints.

To achieve this, I decided to incorporate an ink pad will be placed on the table, inviting viewers to dip their fingers in paint before turning the pages and the book will gradually accumulate the marks of its readers, **becoming a record of their presence**.

Additionally, I plan to pre-mark a few pages with fingerprints myself, to subtly encourage a sense of conformity and **prompt others to participate more willingly**.



I created this kind of texture resembling knife-cut lines by scraping with tree bark.

I used my finger to scoop up the ink from the ink pad.



OUTPUT-POSTERS

Concept

I created a fictional newspaper series called The Traces Times, **compiling old news stories about the impacts of tourism**. In addition, I incorporated a touch of **dark humor** by placing advertisements in the corners of the pages, "discounting" items that cause environmental damage—though their true cost remains extremely high.

Graphic Design

I drew inspiration from **vintage newspaper layouts**, frequently using lines to divide sections and overlapping filled and outlined letters in titles to create a three-dimensional effect. I also employed a variety of typefaces to evoke a retro atmosphere.



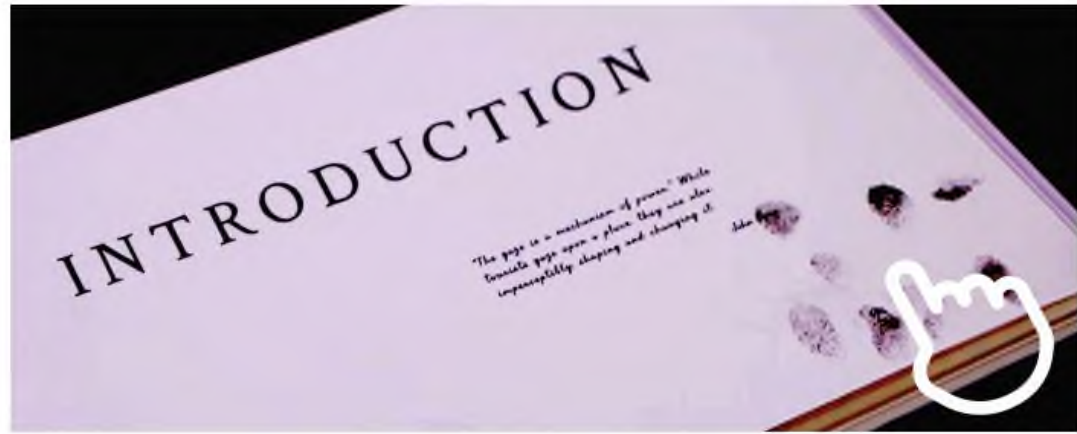
SETTING PROCESS

I chose a rack designed for hanging objects to display the book's pages on a grid, arranging them in a staggered composition. Two display tables were placed in the center to hold the computer and the book.



INTERACTION FLOW

Step1 Dip your hand in ink and leave traces in the book



Step2 Stand before the camera and shape the projected visuals



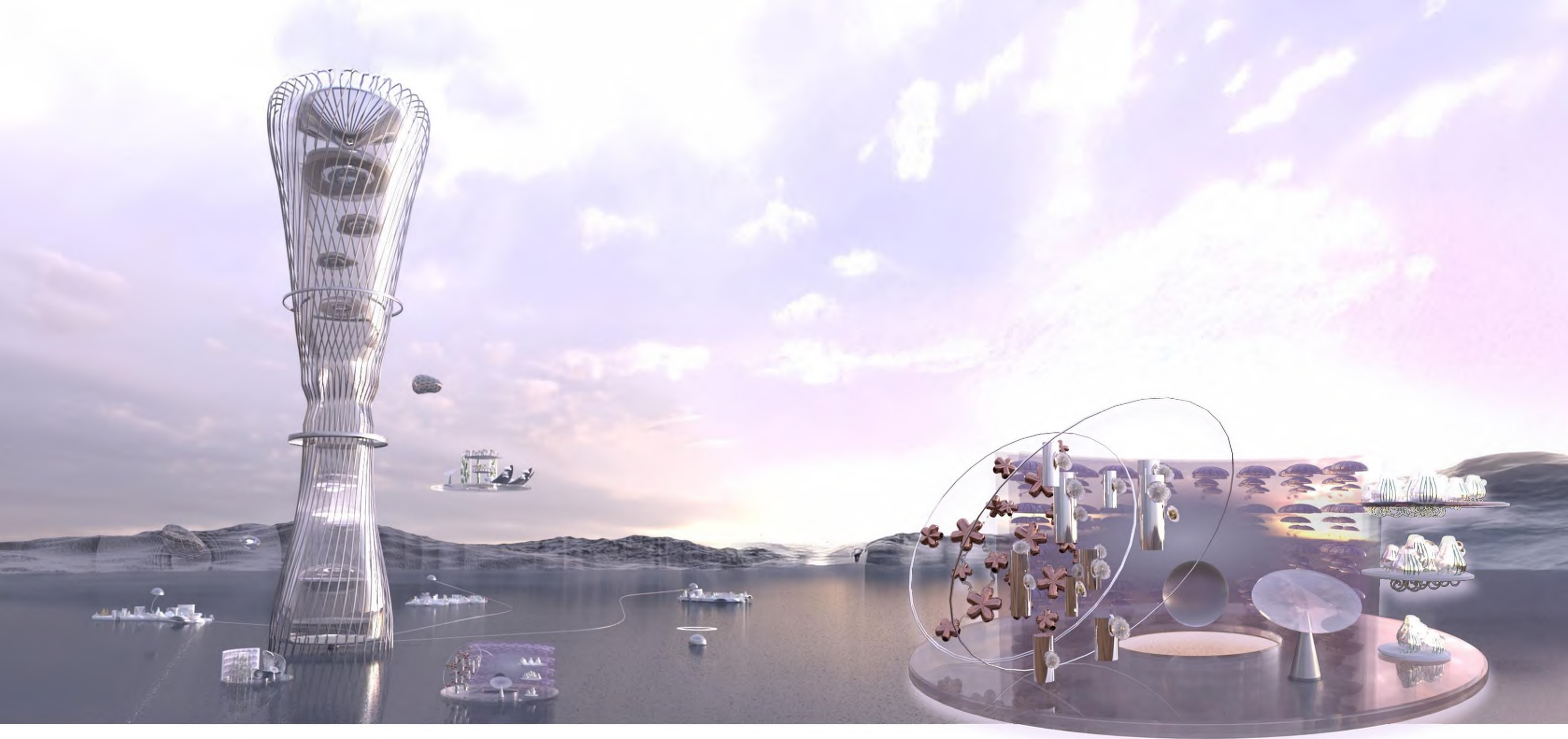
Step3 Freely browse the hanging posters and pages





REFLECTION

This project was my first attempt at combining bookmaking with interactive projection, and it deepened my understanding of **art as a medium that connects people with their environment**. Observing audience reactions made me realize that the act of looking is itself worth examining. The interactive component could still be expanded with more ways to record audience behavior, and the book's content could be further developed for greater narrative richness. Through exploring binding, paper, and interaction, I came to appreciate how readers become active participants as their traces shaping the work as the work responds to them.



EcoSyntax

- Environmental
- Speculation
- Synthetic Ecosystem
- Artificial Biodiversity
- Eco-Futurism

EcoSyntax envisions a futuristic world in 2100, where a fully enclosed, government-funded synthetic ecosystem is created and maintained through advanced technology. Central to this system are engineered plants with specific functions, such as oxygen release, air purification, and food production. Drawing from personal experiences, including the impacts of urbanization on plant diversity, and backed by research into synthetic biology and environmental challenges, the project explores the potential of artificial ecosystems to address future ecological crises.

Through a narrative-driven video inspired by government propaganda, EcoSyntax presents an idealized vision of a synthetic ecosystem while encouraging critical reflection on humanity's control over nature. It raises essential questions about the limits of human intervention, the ethical dilemmas of replacing natural complexity with artificial design, and the long-term consequences of technological dominance in shaping our environment.

Video Link: <https://youtu.be/t-mTNio57As>

INSPIRATION

The Disconnection Between People & Plants

Growing up, I witnessed my grandmother's deep bond with the plants she cultivated. After moving to a compact urban apartment, her garden shrank to a few pots, revealing how urbanization quietly erodes biodiversity and the emotional connection between people and plants.

Later, travelling in Xinjiang exposed me to fragile landscapes where overexploitation pushes ecosystems toward collapse and prompted me to imagine synthetic futures for biodiversity.



INTERVIEW WITH MY GRANDMOTHER

Reflections on Urbanization and Plant Diversity

To gain deeper insights into how **urbanization** compresses planting spaces, reduces plant diversity, and the potential significance of new plants in future lifestyles, I conducted a detailed interview with my grandmother.



Interview Insights

- City life limits planting space, reducing both species variety and emotional engagement.
- Air quality and micro-climate differences make plants harder to grow in urban settings.
- My grandmother believes new technologies might help create resilient plants for future survival.

DESK RESEARCH FINDINGS

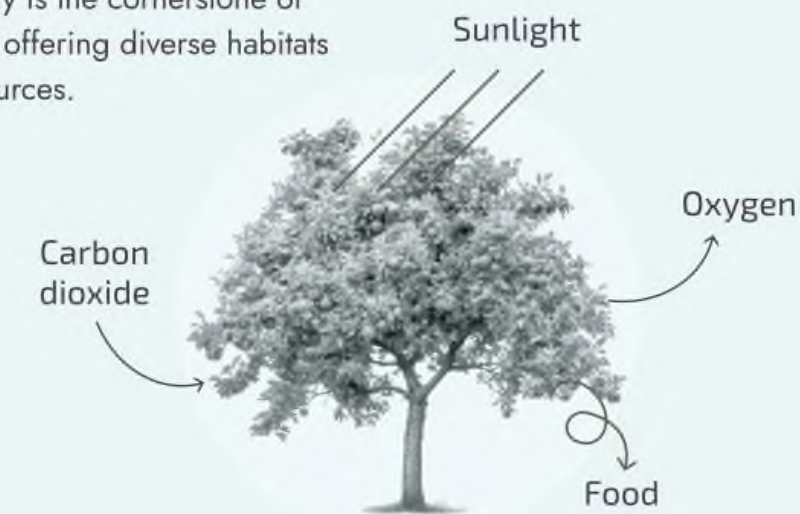
To delve into **the significance of plant diversity and its future potential**, I conducted research from the following four perspectives:

The Importance of Plants to Humanity

The Fundamental Role of Plants in Ecosystems

Plants are primary producers in ecosystems, providing energy for the entire food chain through photosynthesis.

Plant diversity is the cornerstone of biodiversity, offering diverse habitats and food sources.



The Cultural and Social Value of Plants

Agriculture laid the foundation of human civilization and extended its influence into culinary culture, religious rituals, and symbolic meanings.

Gardening activities improve mental health, creating a sense of tranquility and connection with nature, particularly beneficial for the elderly.



The Decline in Plant Diversity

Major Threats



Climate Change



Habitat Loss & Fragmentation



Invasive Species



Overexploitation of Natural Resources

Plant Decline Since 1960s

Since the 1960s, more than 70% of investigated plant species showed declines in nationwide occurrence. Archaeophytes most strongly declined but also native plant species experienced severe declines.⁵



Global Decline in Plant Species

70% ↘

over 70% of investigated plant species have experienced significant declines in nationwide occurrence¹

Threats to Tree Species

1/3 ↘

over 1/3 of global tree species are at risk of extinction, with factors such as deforestation, invasive species, diseases, and climate change being primary contributors²

Climate Change & Plant Extinction Risks

50% ↗

over 50% of the climatically determined geographic range could be lost by 8%, 16%, and 44% of plant species, respectively, increasing their extinction risk³

Plants Vanishing

250 ↗

More than two plant species have gone extinct each year for the past 250 years, a rate 500 times faster than the natural "background extinction rate" without human interference⁴

DESIGN CONCEPT

Core Concept



Design a futuristic scenario for the year 2100

A fully enclosed ecosystem entirely designed and maintained by humans, funded by the government.

Design Features



Artificially Synthesized Plants

Designed for oxygen release, water purification, and food production



Futuristic Narrative Video

A propaganda-style video showcasing EcoSyntax and reflecting on human control over nature

QUESTIONS TO BE EXPLORED

Should human beings control natural ecosystems

What are the **ethical and societal implications** of controlling nature through technology?

Does technological advancement **obscure deeper environmental and cultural issues**?

TECHNOLOGICAL FEASIBILITY

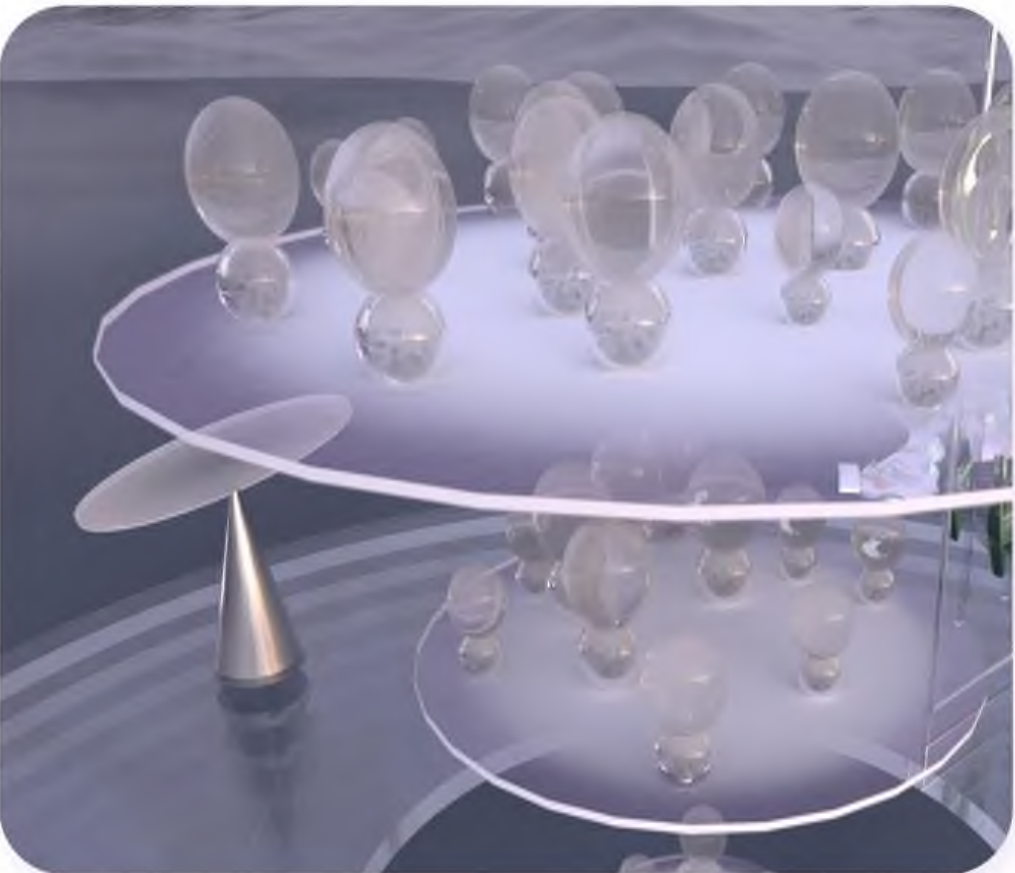
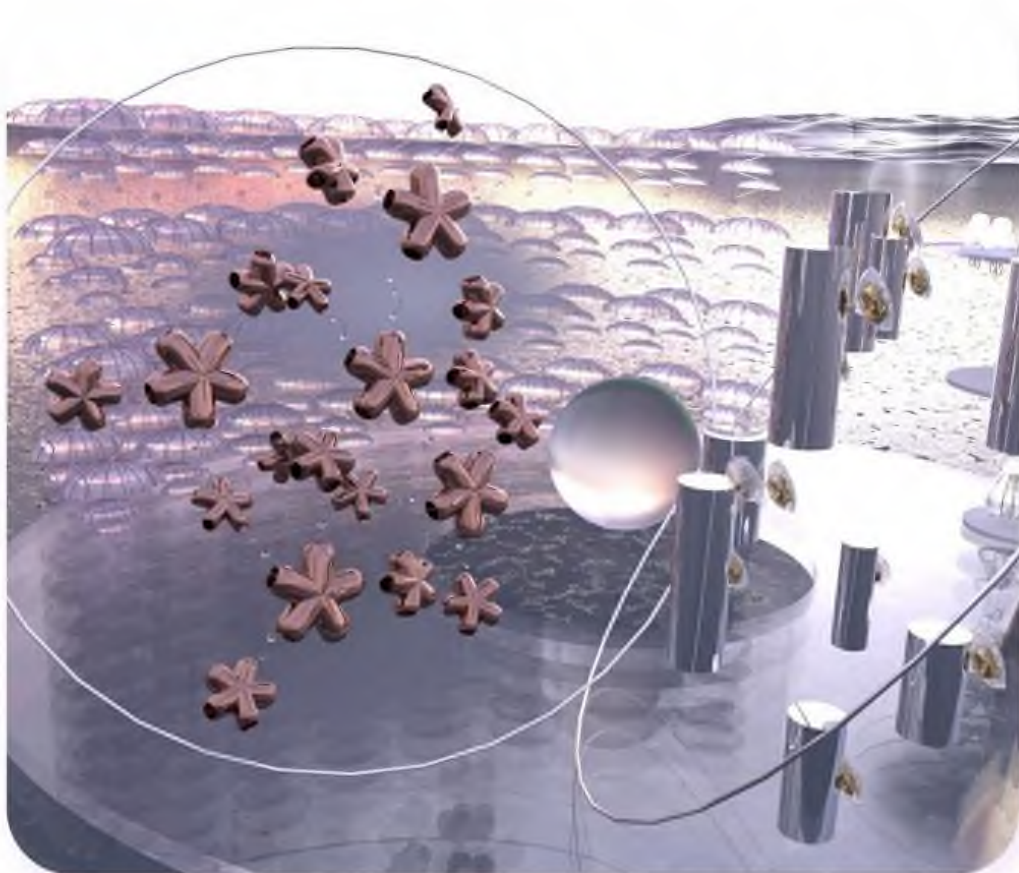
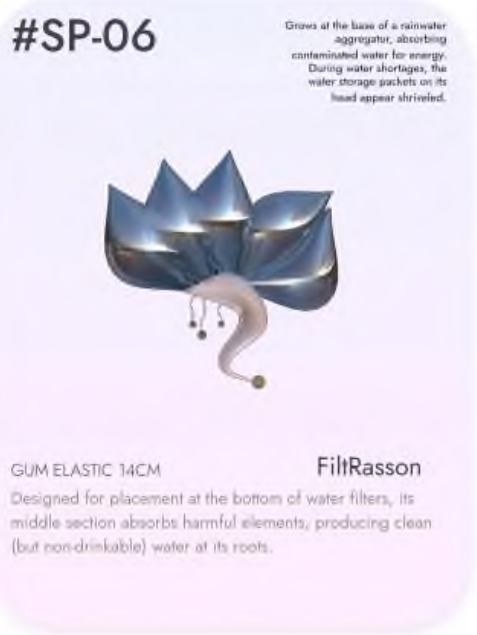
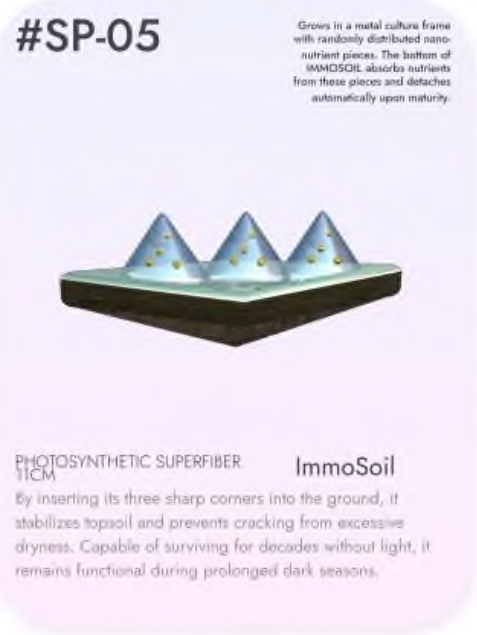
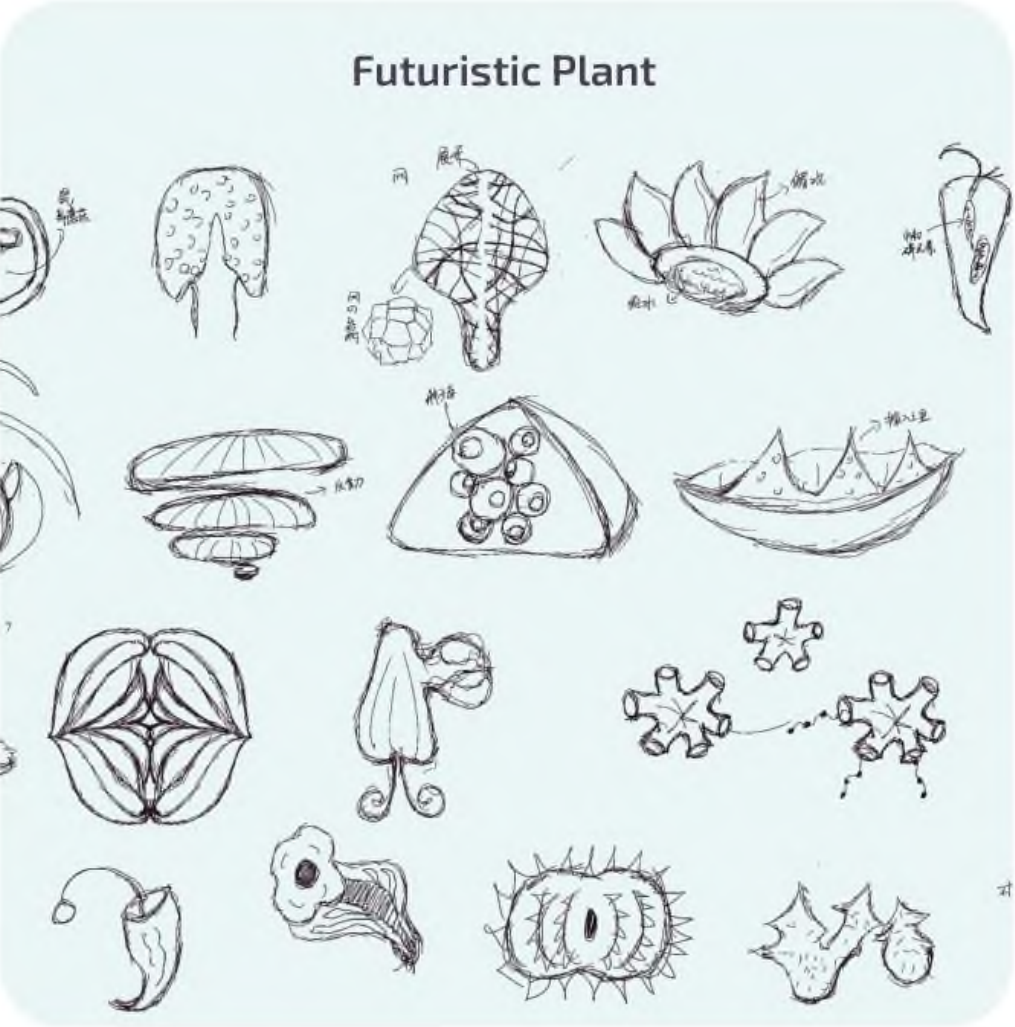
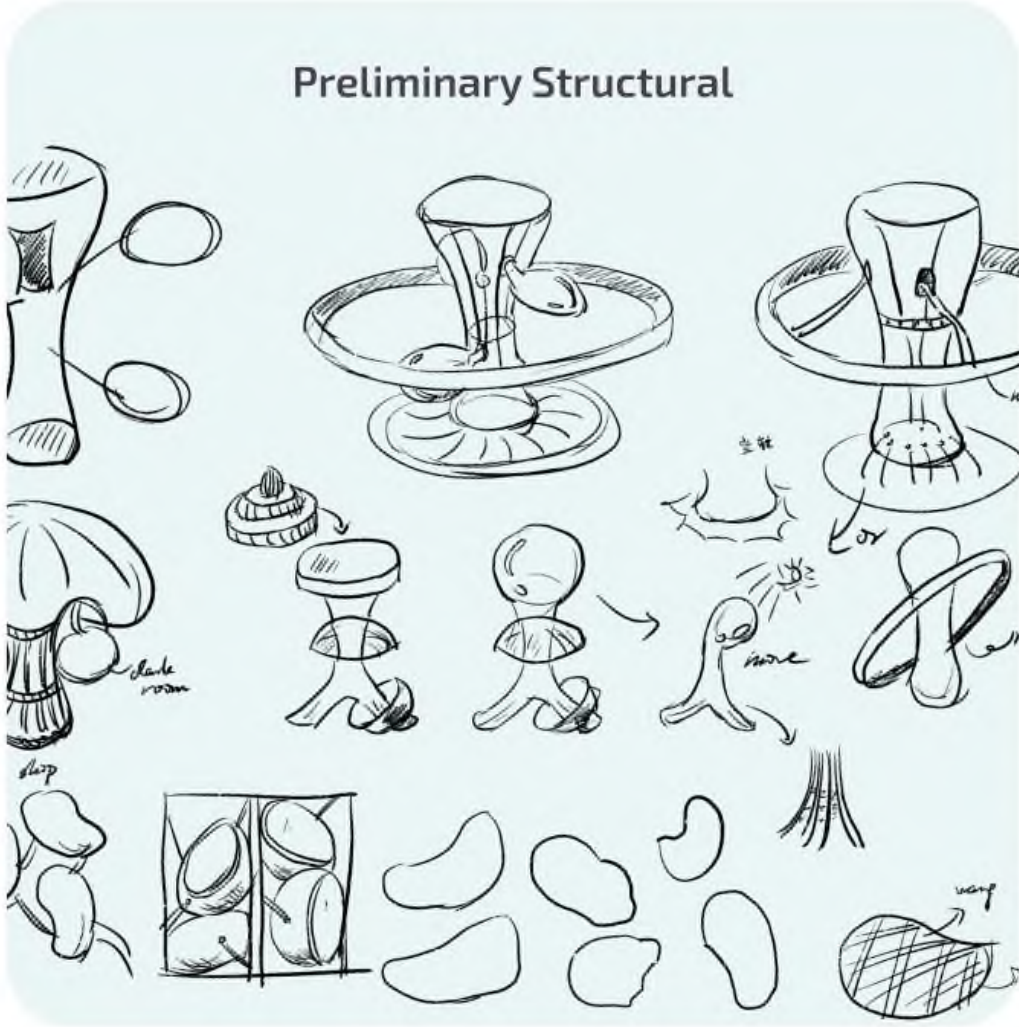
Synthetic Biology: engineering organisms with specific ecological functions
Protein Engineering: customizing biological components for resilience
Plant Synthetic Biology: enhancing traits for future ecosystems

PRESENT-TO-FUTURE MAPPING



DRAFTING

FUTURISTIC PLANT DESIGN

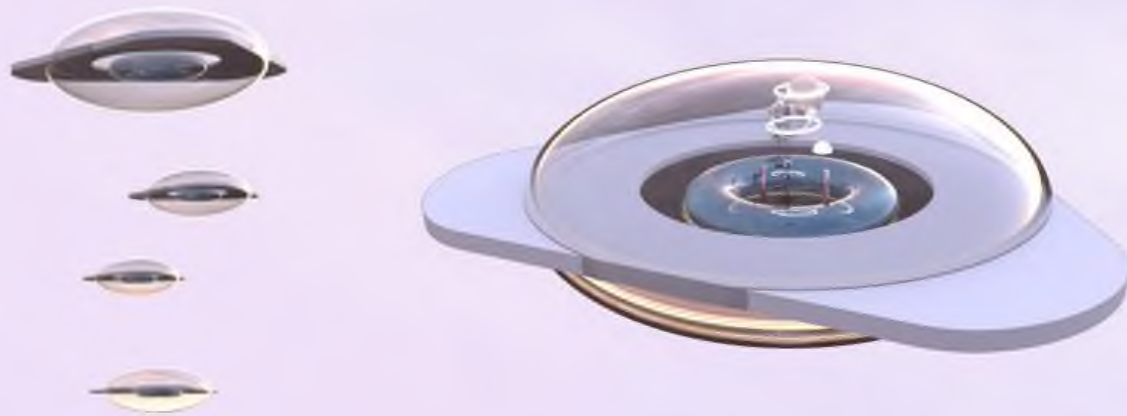


DESIGN

Architectural & Internal structure design

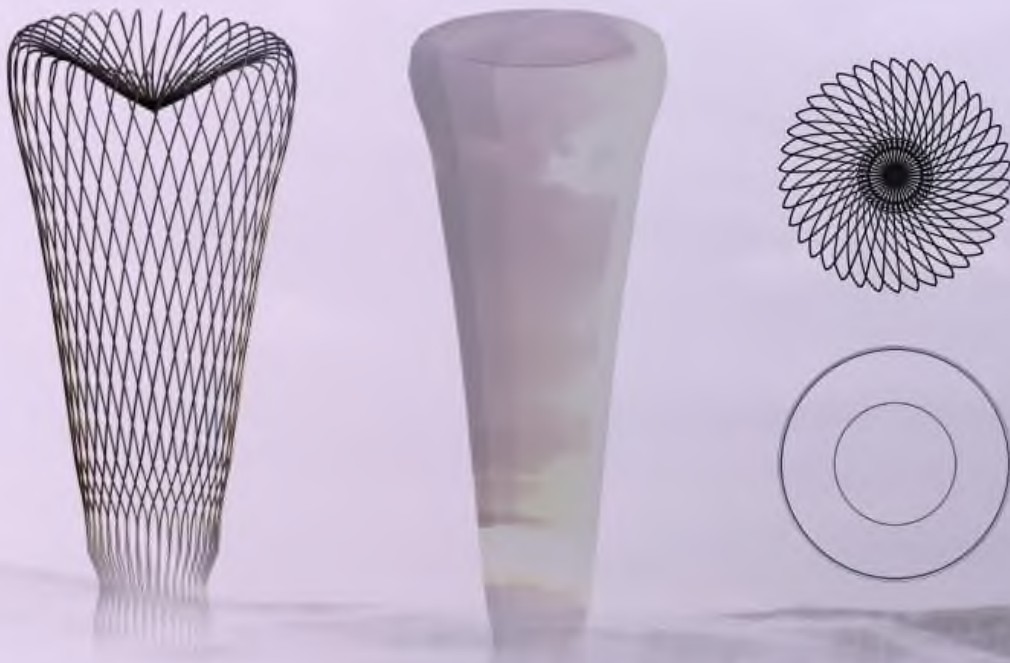
Multilayer Modular Design

The experimental facility consists of multiple levels, each tailored to different ecological needs and functional areas, such as anti-gravity environments and light-free zones. These levels are interconnected through an automated platform and pipeline system.



Anti-Gravity Structure

The building features an hourglass-shaped design with an external mesh structure for protection. Made of high-strength transparent nanomaterials, it filters harmful radiation and effectively prevents external intrusion.



Synthetic Plant Cultivation Platform

A fully automated enclosed system designed for cultivating synthetic plants. Each circular cultivation platform can support up to four plant species, which may have a symbiotic relationship or share the same nutrients.

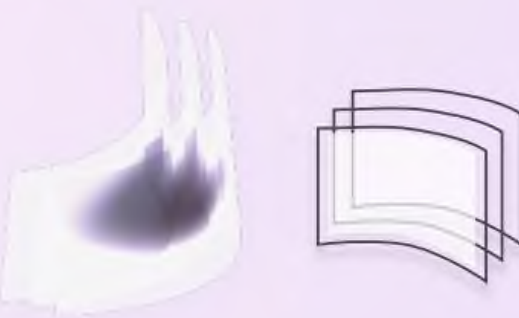
The platform is designed as a mobile, floating circular structure with functions for collecting solar energy and water vapor. It enables precise control of key parameters such as light exposure, temperature, humidity, and CO₂ concentration.



Cultivator Design



Ideal for plants that require extensive exposure to air for optimal growth



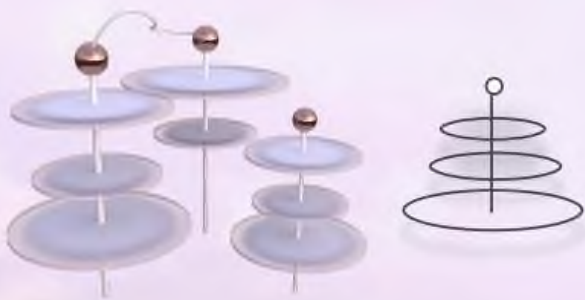
The enclosed vertical cultivation sheet facilitates the large-scale absorption of nutrient solutions by plants



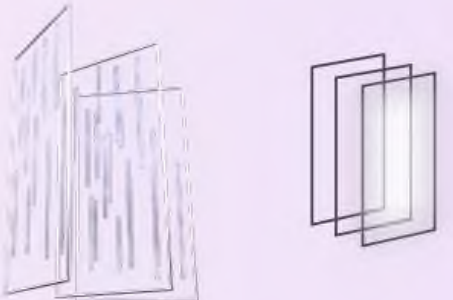
The petal-shaped design allows nutrients to circulate within the cultivator



The ring-shaped cultivator creates a stable growth framework for plants that require a dispersed growth pattern



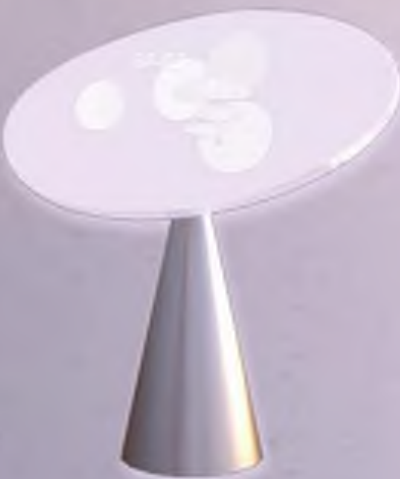
The metal spheres on the cultivator can share info. with those on other cultivators, helping to maintain plant uniformity



The rectangular vertical hollow structure provides a suitable environment for plants that need to grow upside down

Monitoring Dashboard

Scientists can interact with synthetic plants through a virtual reality system, allowing remote monitoring, management, and virtual simulations. This reduces operational risks and resource consumption.



Video Link: <https://youtu.be/t-mTNio57As>